CSC D18 – Computer Graphics – A3 features checklist

For people working in teams, we only need one name + student number to find your submitted work

Student Name (last, first): Student No.: UTORid:

Place an ***X*** in the corresponding column for each features

Feature	*Value*	*Working*	*Partial*	*Not done*	*Test image name*
Antialiasing	2.5	**X**			antialiasing.ppm
Texture Mapping	5	**X**			texture_mapping.ppm
ALS and soft shadows	5	**X**			area_light.ppm
Cool scene	up to 15				
Refraction	up to 10	**X**			refraction.ppm
Multi-threading	2	**X**			(none, see note below)
CSG or Hierarchical	Up to 10				
Normal mapping	5	**X**			normal_mapping.ppm
Alpha or other mappings	2.5	**X**			alpha_mapping.ppm
Depth of field	up to 7				
Photon mapping	up to 12				
Dispersion	5				
RT Acceleration	up to 12				
Ray Marching	up to 7				

Provide a list of scripts that must be run to generate the test images noted above:

Run one script to build and render everything (it can be run from any directory):

```
demo/scripts/run_all.sh
```

It runs the scripts below in parallel. Each one compiles its own binary into demo/bin/, picking the scene and features with -D flags (see the top of buildscene.c), then renders a 1024x1024 image into demo/renderings/. Each script's output is saved in demo/logs/. The whole run takes about a minute on a multi-core machine.

```
demo/scripts/base.sh              base.ppm             no A3 features, for comparison
demo/scripts/antialiasing.sh      antialiasing.ppm     base scene, antialiasing on
demo/scripts/texture_mapping.sh   texture_mapping.ppm  -DTEXTURES
demo/scripts/normal_mapping.sh    normal_mapping.ppm   -DTEXTURES -DNORMALS
demo/scripts/area_light.sh        area_light.ppm       -DALS
demo/scripts/alpha_mapping.sh     alpha_mapping.ppm    -DALPHA
demo/scripts/refraction.sh        refraction.ppm       -DSCENE_REFRACTION (separate scene)
```

Refraction: a glass marble (n = 1.9) inside a larger glass sphere (n = 1.5). Each ray keeps a stack of the objects it is inside, so every surface uses the correct indices on both sides (air -> glass -> marble -> glass -> air). Under total internal reflection, the light that cannot refract out is reflected instead.

Multi-threading

Rendering always runs on a thread pool (one worker per CPU core, each taking the next unrendered row), so every image above uses it. This is hard to show in an image, so you'll have to take my word on this one: after adding the thread pool, a scene that took 10-11 seconds to render took only 2-3 seconds.